

Assignment 02

2102024

chapter 08

Building a decision Tree [C4.5]

$$\begin{aligned} \text{Info}(D) &= - \left[\frac{10}{20} \log_2 \frac{10}{20} + \frac{10}{20} \log_2 \frac{10}{20} \right] \\ &= - \left(-\frac{1}{2} - \frac{1}{2} \right) \\ &= 1 \end{aligned}$$

Genders

$$\begin{aligned} \text{Info}(D, \text{Gender}) &= \frac{10}{20} \left[-\frac{6}{10} \log_2 \frac{6}{10} - \frac{4}{10} \log_2 \frac{4}{10} \right] + \frac{10}{20} \left[-\frac{4}{10} \log_2 \frac{4}{10} - \frac{6}{10} \log_2 \frac{6}{10} \right] \\ &= 0.97095 \end{aligned}$$

Car

$$\begin{aligned} \text{Info gain}(\text{Gender}) &= 1 - 0.97095 \\ &= 0.02905 \end{aligned}$$

$$\text{split-info}(\text{Gender}) = - \frac{10}{20} \log_2 \frac{10}{20}$$

$$\left[- \frac{10}{20} \log_2 \frac{10}{20} \right] = 1$$

$$\text{Gain ratio}(\text{GIPA}) = \frac{0.02905}{1} = 0.02905$$

ID	Gender	Car	Shirt class
1	male	Fam	small c1
2	male	Spor.	med c1
3	male	Spo.	med c1
4	Male	Spo.	large c1
5	Male	Spo.	Ex. lar c1
6	Male	Spo.	Ex. lar c1
7	Fem	Spo.	small c1
8	Female	Spo.	small c1
9	Fem	Spo.	med c1
10	Fem	Lux	large c1
11	male	Fam	large c2
12	male	Fam	Ex. lar c2
13	male	Fam	med c2
14	Male	Lux	Ex. larg. c2
15	Fem	Lux	small c2
16	Fem	Lux	small c2
17	Fem	Lux	med c2
18	Fem	Lux	med c2
19	Fem	Lux	med c2
20	Fem	Lux	large c2

Car

$$\begin{aligned} \text{Info}(D, \text{Car}) &= \frac{4}{20} \left[-\frac{1}{4} \log_2 \left(\frac{1}{4} \right) - \frac{3}{4} \log_2 \left(\frac{3}{4} \right) \right] \\ &+ \frac{8}{20} \left[-\frac{8}{8} \log_2 \left(\frac{8}{8} \right) - \frac{0}{8} \log_2 \left(\frac{0}{8} \right) \right] \\ &+ \frac{8}{20} \left[-\frac{1}{8} \log_2 \left(\frac{1}{8} \right) - \frac{7}{8} \log_2 \left(\frac{7}{8} \right) \right] \\ &= 0.37968 \end{aligned}$$

$$\text{Info gain}(\text{Car}) = 1 - 0.37968 = 0.62032$$

$$\begin{aligned} \text{split info}(\text{Car}) &= -\frac{1}{20} \log_2 \frac{1}{20} - \frac{8}{20} \log_2 \frac{8}{20} - \frac{8}{20} \log_2 \frac{8}{20} \\ &= 1.521928 \end{aligned}$$

$$\text{Gain ratio}(\text{Car}) = \frac{0.37968}{1.521928} = 0.249473$$

Shirt

$$\begin{aligned} \text{Info}(D, \text{Shirt}) &= \frac{5}{20} \left[-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \right] \\ &+ \frac{7}{20} \left[-\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} \right] \\ &+ \frac{4}{20} \left[-\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \right] \\ &+ \frac{4}{20} \left[-\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \right] \end{aligned}$$

$$= 0.98757$$

$$\text{Info gain (shirt)} = 1 - 0.98757$$

$$= 0.01243$$

$$\text{split info (shirt)} = -\frac{5}{20} \log_2 \frac{5}{20} - \frac{7}{20} \log_2 \frac{7}{20} - \frac{4}{20} \log_2 \frac{4}{20} \times 2$$

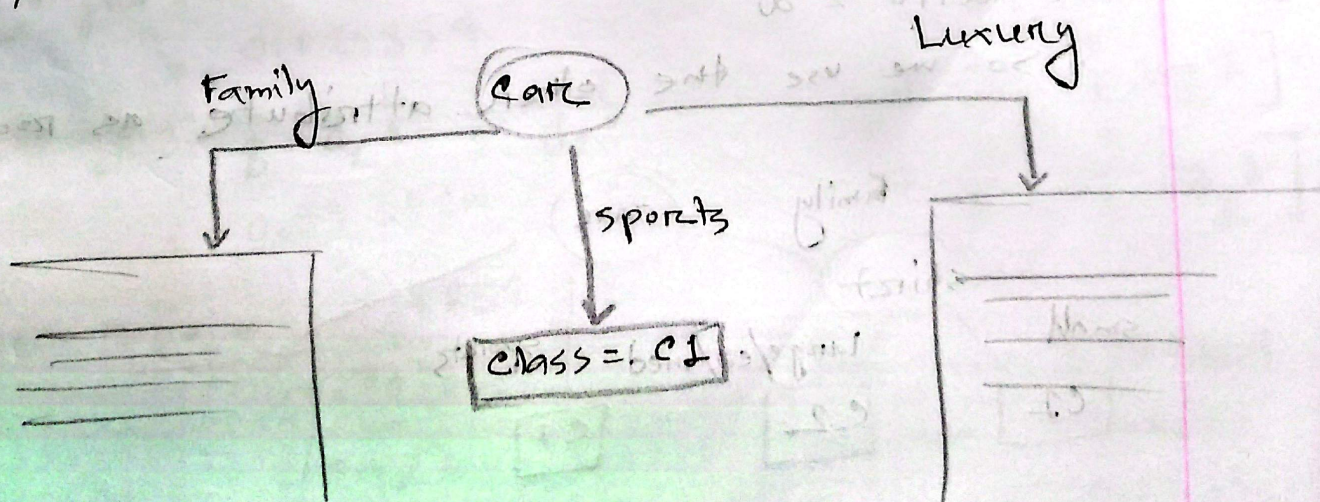
$$= 1.95887$$

$$\text{Gain ratio (shirt)} = \frac{0.01243}{1.95887} = 0.006345$$

Here, gain ratios are

Attribute	Gain ratio
Gender	0.02905
Car	0.249473
Shirt	0.006345

Here car has the greater gain ratio. So it will be our root node.



If car type is Family, then,

ID	Gender	shirt	Class
1	Male	Small	c1
11	Male	Large	c2
12	Male	Ex. La.	c2
13	Male	Medium	c2

$$\text{Info}(\text{Family}) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 1$$

For gender

$$\text{Info}(\text{Family, Gen}) = \frac{4}{4} \left(-\frac{4}{4} \log_2 \frac{4}{4} \right) + \frac{0}{4}$$

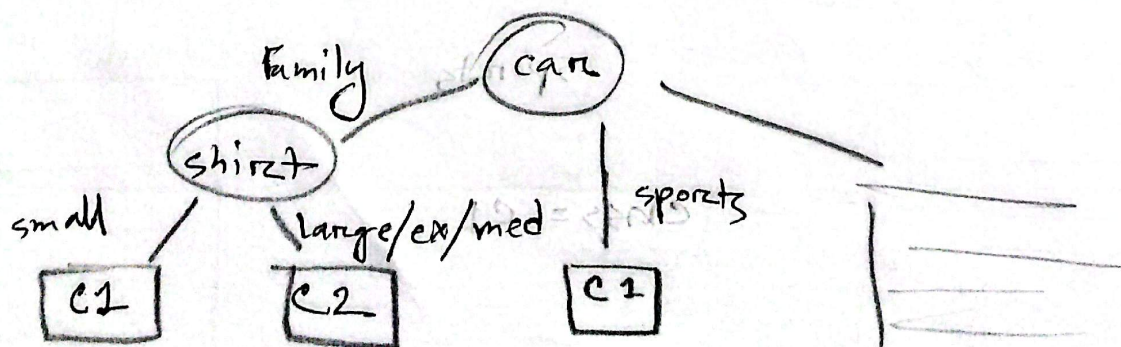
$$= 0$$

$$\text{Info gain} = 1 - 0 = 1$$

$$\text{split info} = -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} = 0$$

$$\text{Gain ratio} = \infty$$

So we use the other attribute as root



If car type is luxury, then,

ID	Gender	Shirt	Class
10	Female	Large	c1
14	Male	Ex L	c2
15	Female	S	c2
16	"	S	c2
17	"	M	c2
18	"	M	c2
19	"	M	c2
20	"	L	c2

For gender

$$\text{Info (Luxury)} = -\frac{1}{8} \log_2 \frac{1}{8} - \frac{7}{8} \log_2 \frac{7}{8} \\ = 0.543564$$

$$\text{Info (Gen)} = -\frac{1}{8} \left(-\frac{1}{2} \log_2 \frac{1}{2} \right) \\ + \frac{7}{8} \left(-\frac{1}{7} \log_2 \frac{1}{7} - \frac{6}{7} \log_2 \frac{6}{7} \right) \\ = 0.51771$$

$$\text{Info gain} = 0.543564 - 0.51771 \\ = 0.025854$$

$$\text{split info} = -\frac{1}{8} \log_2 \frac{1}{8} - \frac{7}{8} \log_2 \frac{7}{8} \\ = 0.543564$$

$$\text{Gain ratio} = \frac{0.025854}{0.543564} \\ = 0.0475638$$

For shirt

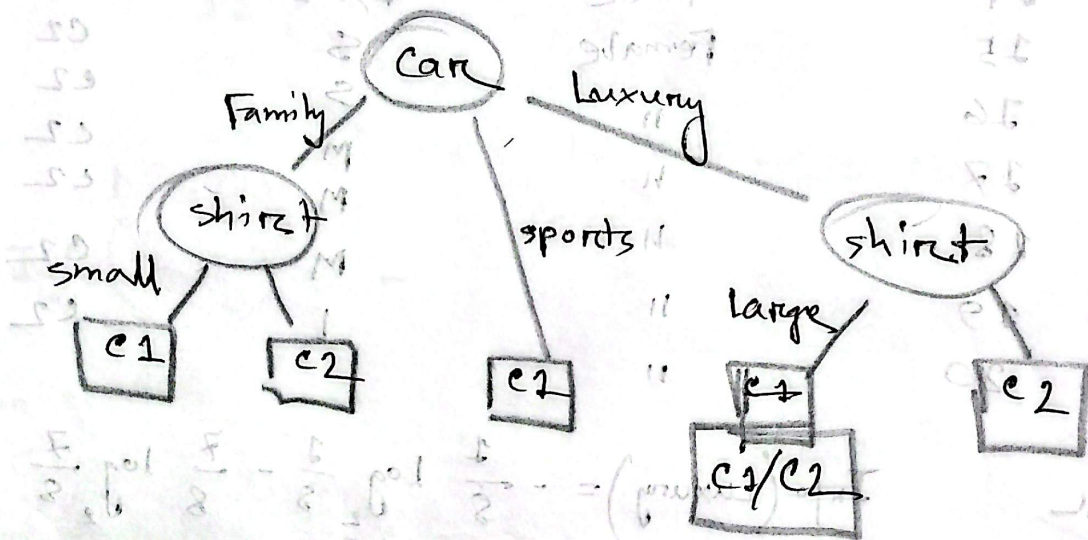
$$\text{Info} = \frac{2}{8} \left[-\frac{1}{2} \log_2 \frac{1}{2} \times 2 \right] \\ + \frac{1}{8} \left[-\frac{1}{2} \log_2 \frac{1}{2} \right] \\ + \frac{2}{8} \left[-\frac{2}{2} \log_2 \frac{1}{2} \right] \\ + \frac{3}{8} \left[-\frac{3}{3} \log_2 \frac{1}{3} \right] \\ = 0.25$$

$$\text{Info gain} = 0.293564$$

$$\text{split info} = 1.905639$$

$$\text{Gain ratio} = \frac{0.293564}{1.905639} \\ = 0.15405$$

Here gain ratio of shirt is greater. So it will be the root here. So our final decision tree will be,



Here, c1 and c2 both has the same attributes. So it can be either c1/c2.

$$\left[5 \times \frac{1}{5} \log_2 \frac{1}{5} + 5 \times \frac{1}{5} \log_2 \frac{1}{5} \right] \frac{1}{10} = 0.47$$

$$\left[4 \times \frac{1}{4} \log_2 \frac{1}{4} + 4 \times \frac{1}{4} \log_2 \frac{1}{4} \right] \frac{1}{8} = 0.5$$

$$\left[6 \times \frac{1}{6} \log_2 \frac{1}{6} + 6 \times \frac{1}{6} \log_2 \frac{1}{6} \right] \frac{1}{12} = 0.58$$

$$\left[5 \times \frac{1}{5} \log_2 \frac{1}{5} + 5 \times \frac{1}{5} \log_2 \frac{1}{5} \right] \frac{1}{10} = 0.47$$

$$0.5$$

$$0.47 \times 0.5 = 0.235$$

$$0.5 \times 0.5 = 0.25$$

$$0.58 \times 0.5 = 0.29$$

$$0.47 \times 0.5 = 0.235$$

$$0.47 \times 0.5 = 0.235$$

$$0.5 \times 0.5 = 0.25$$

$$0.58 \times 0.5 = 0.29$$

$$0.47 \times 0.5 = 0.235$$

$$0.235 \times 0.5 = 0.1175$$

$$0.25 \times 0.5 = 0.125$$

$$0.29 \times 0.5 = 0.145$$

$$0.235 \times 0.5 = 0.1175$$

Chapter 08

Building a Decision Tree [CART]

$$\text{Gini Index}(T) = 1 - \left(\frac{10}{20}\right)^2 - \left(\frac{10}{20}\right)^2$$

$$= \frac{1}{2} = 0.5$$

For car

car	c1	c2
Family	1	3
sports	8	0
luxury	1	7

$$\text{Gini}(T, \{Family, sports\}) = 1 - \left(\frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2$$

$$= 0.375$$

$$\text{Gini}(T, \{sports\}) = 1 - \left(\frac{8}{8}\right)^2 - \left(\frac{0}{8}\right)^2$$

$$= 0$$

$$\text{Gini}(T, \{lux\}) = 1 - \left(\frac{1}{8}\right)^2 - \left(\frac{7}{8}\right)^2$$

$$= 0.21875$$

$$\text{Gini}(T, \{Family, sports\}) = 1 - \left(\frac{9}{12}\right)^2 - \left(\frac{3}{12}\right)^2 = 0.375$$

$$\text{Gini}(T, \{Family, sports, luxury\}) = \frac{12}{20} \times 0.375 + \frac{8}{20} \times 0.375$$

$$\text{Gini}(T, \{Family, luxury\}) = 1 - \left(\frac{2}{12}\right)^2 - \left(\frac{10}{12}\right)^2 = 0.27778$$

$$\text{Gini}(T, \{Family, luxury, sports\}) = 1 - \left(\frac{2}{12}\right)^2 - \left(\frac{10}{12}\right)^2$$

$$\text{Gini}(T, \{sports, luxury\}) = 1 - \left(\frac{2}{16}\right)^2 - \left(\frac{7}{16}\right)^2 = 0.4921875$$

$$\text{Gini}(T, \{sports, luxury, Family\}) = 1 - \left(\frac{9}{16}\right)^2 - \left(\frac{7}{16}\right)^2$$

$$\textcircled{*} \text{ Gini}(T, \{(Family, luxury), sports\}) = \frac{12}{20} \times 0.27778 + \frac{8}{20} \times 0$$

$$= 0.16668$$

$$\text{Gini}(T, \{(Family, sports, luxury), Family\}) = \frac{16}{20} \times 0.4921875$$

$$+ \frac{4}{20} \times 0.375 = 0.46875$$

$$\text{Gini}(T, \{(Family, sports), luxury\}) = \frac{12}{20} \times 0.375 + \frac{8}{20} \times 0.375$$

$$= 0.375$$

For gender

$\Delta \text{Gini}(\text{car})$ will depend on the best splitting which has minimum gini index. $\{(Family, luxury), sports\}$ has the lowest gini index

$$\therefore \Delta \text{Gini}(\text{car}) = 0.5 - 0.16668 = 0.33332$$

For gender

$$\text{Gini}(T, \{\text{male}\}) = 1 - \left(\frac{6}{10}\right)^2 - \left(\frac{4}{10}\right)^2 = 0.48$$

$$\text{Gini}(T, \{\text{female}\}) = 1 - \left(\frac{4}{10}\right)^2 - \left(\frac{6}{10}\right)^2 = 0.48$$

$$\text{Gini}(T, \{\text{male, female}\}) = \frac{10}{20} \times 0.48 + \frac{10}{20} \times 0.48$$

$$= 0.48$$

$$\therefore \Delta \text{Gini}(\text{Gender}) = 0.5 - 0.48 = 0.02$$

For shirt

shirt	c1	c2
small	3	2
medium	3	4
large	2	2
Extra large	2	2

$$Gini(T, \{small\}) = 1 - \left(\frac{3}{5}\right)^2 - \left(\frac{2}{5}\right)^2 = 0.48$$

$$Gini(T, \{medium\}) = 0.489796$$

$$Gini(T, \{large\}) = 0.5$$

$$Gini(T, \{Extra Large\}) = 0.5$$

$$Gini(T, \{small, medium\}) = 0.5$$

$$Gini(T, \{small, large\}) = 0.4938$$

$$Gini(T, \{small, XL\}) = 0.4938$$

$$Gini(T, \{med, large\}) = 0.4958$$

$$Gini(T, \{med, XL\}) = 0.4958$$

$$Gini(T, \{large, XL\}) = 0.5$$

$$Gini(T, \{small, med, large\}) = 0.5$$

$$Gini(T, \{small, med, XL\}) = 0.5$$

$$Gini(T, \{small, large, XL\}) = 0.49704$$

$$Gini(T, \{med, large, XL\}) = 0.49728$$

$$* \text{Gini}(T, \{(small), (med, large, XL)\}) = \frac{5}{20} \times 0.48 + \frac{15}{20} \times 0.49778 = 0.493335$$

$$\text{Gini}(T, \{(medium), (small, L, XL)\}) = 0.49845046$$

$$\text{Gini}(T, \{(large), (small, M, XL)\}) = 0.5$$

$$\text{Gini}(T, \{(XL), (small, M, L)\}) = 0.5$$

$$\text{Gini}(T, \{(small, med), (L, XL)\}) = 0.5$$

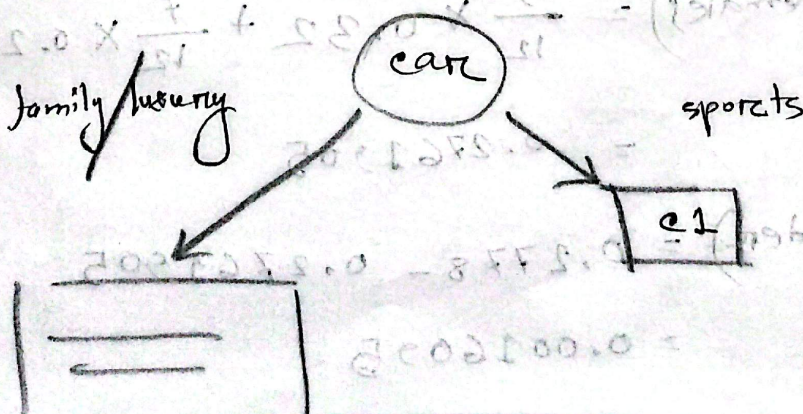
$$\text{Gini}(T, \{(S, L), (M, XL)\}) = 0.4949$$

$$\text{Gini}(T, \{(small, XL), (med, L)\}) = 0.4949$$

Here, $\{(small), (med, large, XL)\}$ has the lowest gini index

$$\therefore \Delta \text{Gini}(\text{shirt}) = 0.5 - 0.493335 = 0.006665$$

Here, car has the greater value of ΔG , hence it is the best splitting sub set



If we consider "family/luxury" car,

ID	Gender	Car	Shirt	Class
1	Male	Fam	S	c1
10	Female	Lux	L	c1
11	Male	Fam	L	c2
12	Male	Fam	XL	c2
13	Male	Fam	M	c2
14	Male	Lux	XL	c2
15	Female	Lux	S	c2
16	"	Lux	S	c2
17	"	Lux	M	c2
18	"	Lux	M	c2
19	"	Lux	M	c2
20	"	Lux	L	c2

For gender, Gini index = $1 - \left(\frac{2}{12}\right)^2 - \left(\frac{10}{12}\right)^2 = 0.2778$

$Gini(T, \{male\}) = 1 - \left(\frac{1}{5}\right)^2 - \left(\frac{4}{5}\right)^2 = 0.32$

$Gini(T, \{female\}) = 1 - \left(\frac{1}{7}\right)^2 - \left(\frac{6}{7}\right)^2 = 0.244898$

$Gini(T, \{male, female\}) = \frac{5}{12} \times 0.32 + \frac{7}{12} \times 0.244898$
 $= 0.2761905$

$\therefore \Delta Gini(gender) = 0.2778 - 0.2761905$
 $= 0.0016095$

For car,

$$\text{Gini}(T, \{Fam\}) = 1 - \left(\frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2 = 0.375$$

$$\text{Gini}(T, \{Lux\}) = 1 - \left(\frac{1}{8}\right)^2 - \left(\frac{7}{8}\right)^2 = 0.21875$$

$$\begin{aligned} \text{Gini}(T, \{Fam, Lux\}) &= \frac{4}{12} \times 0.375 + \frac{8}{12} \times 0.21875 \\ &= 0.270833 \end{aligned}$$

$$\therefore \Delta \text{Gini}(\text{car}) = 0.2778 - 0.270833$$

$$= 0.006967$$

For shirt

shirt	c1	c2
S	1 1	2
M	0 4	0 4
L	1 1	2
XL	0 2	0 2

$$\text{Gini}(T, \{S\}) = 1 - 0.4444$$

$$\text{Gini}(T, \{M\}) = 0$$

$$\text{Gini}(T, \{L\}) = 0.4444$$

$$\text{Gini}(T, \{XL\}) = 0$$

$$\text{Gini}(T, \{S, M\}) = 0.2445$$

$$\text{Gini}(T, \{S, M, L\}) = 0.32$$

$$\text{Gini}(T, \{S, L\}) = 0.4444$$

$$\text{Gini}(T, \{S, M, XL\}) = 0.19753$$

$$\text{Gini}(T, \{S, XL\}) = 0.32$$

$$\text{Gini}(T, \{M, L\}) = 0.2445$$

$$\text{Gini}(T, \{S, L, XL\}) = 0.375$$

$$\text{Gini}(T, \{M, XL\}) = 0$$

$$\text{Gini}(T, \{M, L, XL\}) = 0.1975$$

$$\text{Gini}(T, \{L, XL\}) = 0.32$$

$$\text{Gini}(T, \{(S), (M, L, XL)\}) = 0.259225$$

$$\text{Gini}(T, \{(M), (S, L, XL)\}) = 0.25$$

$$\text{Gini}(T, \{(L), (S, M, XL)\}) = 0.2592975$$

$$\text{Gini}(T, \{(XL), (S, M, L)\}) = 0.26667$$

$$\text{Gini}(T, \{(S, M), (L, XL)\}) = 0.27625$$

$$\text{Gini}(T, \{(S, L), (M, XL)\}) = 0.2222$$

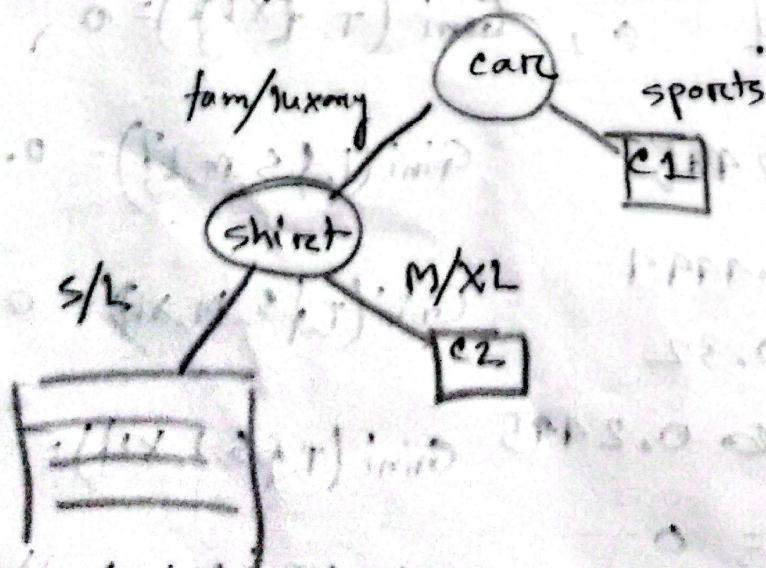
$$\text{Gini}(T, \{(S, XL), (M, L)\}) = 0.27625$$

Here, $\{(S, L), (M, XL)\}$ has the lowest gini index.

$$\Delta \text{Gini}(\text{shirt}) = 0.2778 - 0.2222$$

$$= 0.0556$$

Here, shirt has greater value of ΔGini , so



Here, for shirt, S/L,

ID	Gender	Car	shirt	class
1	male	Fam	S	c1
10	Female	Lux	L	c1
11	Male	Fam	L	c2
25	Female	Lux	S	c2
26	Female	Lux	S	c2
20	Female	Lux	L	c2

$$\therefore \text{Gini index} = 1 - \left(\frac{2}{6}\right)^2 - \left(\frac{4}{6}\right)^2 = 0.4444$$

For gender,

$$\text{Gini}(T, \{ \text{male} \}) = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 0.5$$

$$\text{Gini}(T, \{ \text{female} \}) = 1 - \left(\frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2 = 0.375$$

$$\text{Gini}(T, \{ \text{male, female} \}) = \frac{2}{6} \times 0.5 + \frac{4}{6} \times 0.375$$

$$= 0.416667$$

$$\Delta \text{Gini}(\text{gender}) = 0.4444 - 0.416667$$

$$= 0.027773$$

For car,

$$\text{Gini}(T, \{Fam\}) = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 0.5$$

$$\text{Gini}(T, \{Lux\}) = 1 - \left(\frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2 = 0.375$$

$$\text{Gini}(T, \{Fam, Lux\}) = \frac{2}{6} \times 0.5 + \frac{4}{6} \times 0.375$$

$$\Delta \text{Gini}(\text{car}) = 0.027773$$

For shirt,

$$\text{Gini}(T, \{S\}) = 1 - \left(\frac{1}{3}\right)^2 - \left(\frac{2}{3}\right)^2 = 0.4444$$

$$\text{Gini}(T, \{L\}) = 1 - \left(\frac{1}{3}\right)^2 - \left(\frac{2}{3}\right)^2 = 0.4444$$

$$\text{Gini}(T, \{S, L\}) = \frac{3}{6} \times 0.4444 + \frac{3}{6} \times 0.4444 = 0.4444$$

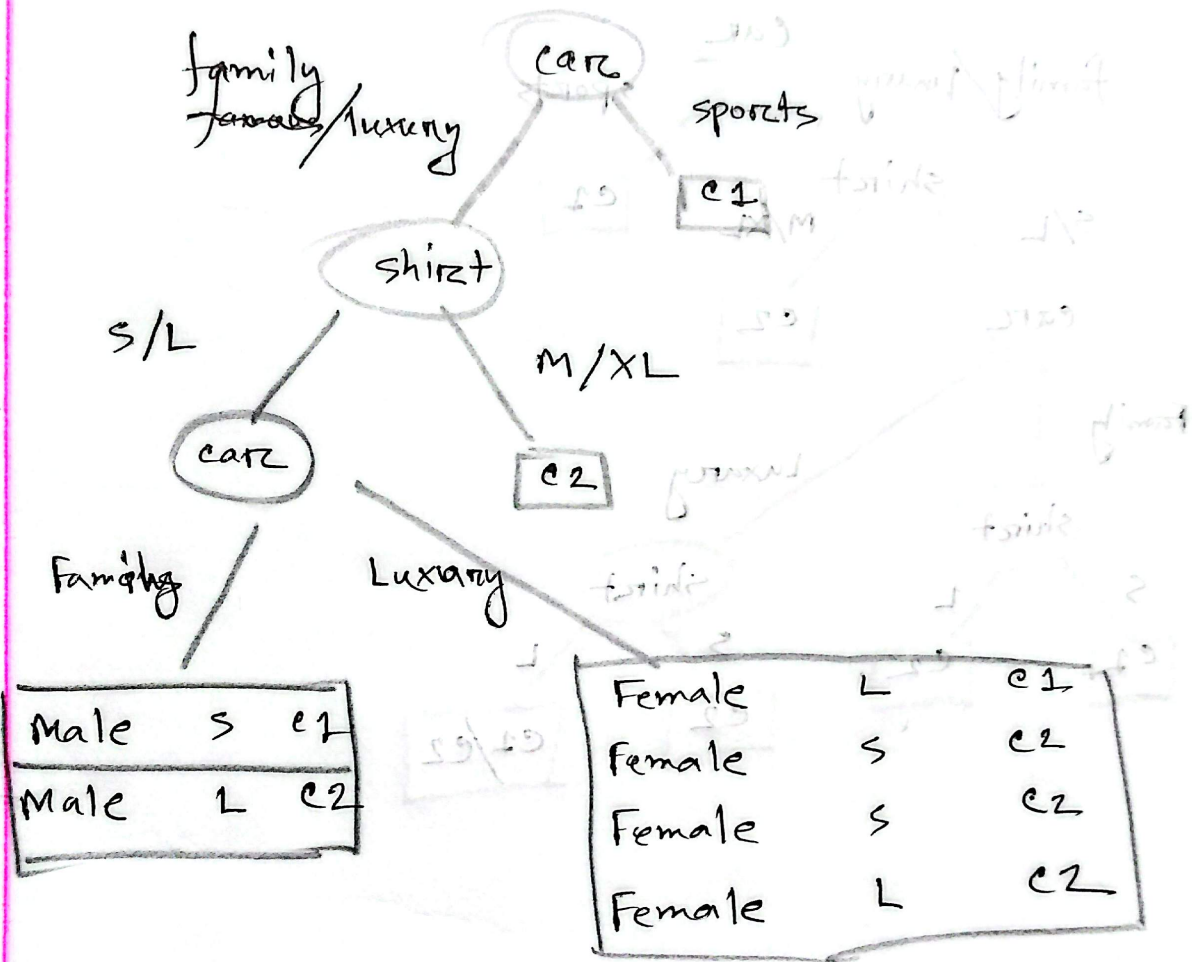
$$\Delta \text{Gini}(\text{shirt}) = 0.4444 - 0.4444$$

$$= 0$$

Here, $\Delta G(\text{car})$ or $\Delta G(\text{Gender})$ has the same value, so we can take either.

Let's consider cars.

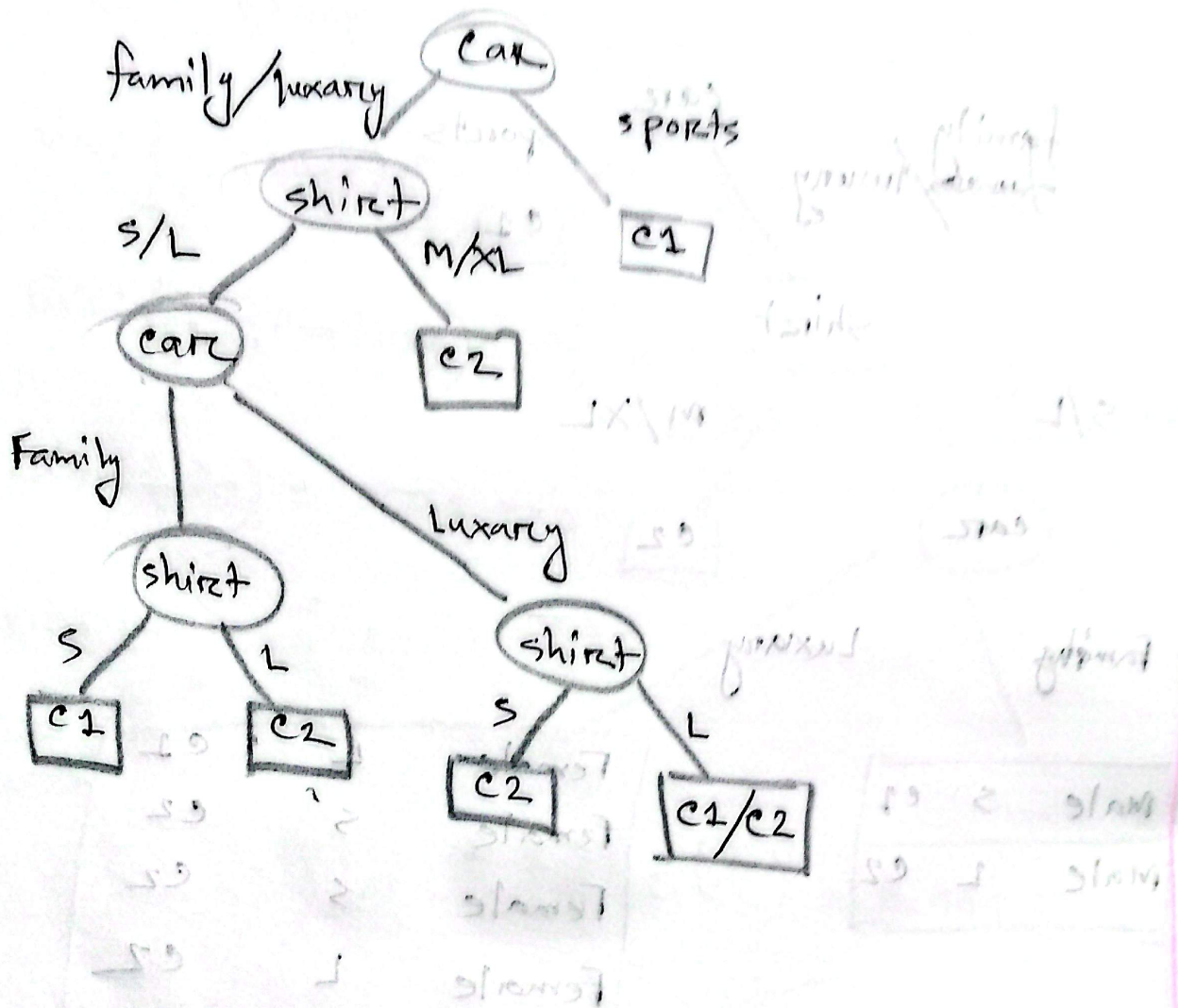
Our decision tree will be, given below



As there are attributes with all same values, so we can simply keep another one as root and form our final tree.

In this case, for female L, there can be either e1/e2.

Finally our decision tree will be,



As there are attributes like all, some, none, etc. we can simply keep another one as leaf node for each attribute. In this case for female, we can be either c1/c2.